

1.2: Matrices

1. Introduction to Matrices

Matrix: A matrix is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns. The entry in row i , column j is denoted a_{ij} .

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Special Matrices

- **Identity matrix I :** Square matrix with 1s on the diagonal, 0s elsewhere. $AI = IA = A$.
- **Zero matrix O :** All entries are 0. $A + O = A$.
- **Transpose A^T :** Rows and columns swapped. $(A^T)_{ij} = A_{ji}$.
- **Symmetric matrix:** $A^T = A$.

2. Matrix Operations

Addition and Scalar Multiplication

Matrices must be the same size to add. Add corresponding entries. Scalar multiplication multiplies every entry:

$$(A + B)_{ij} = a_{ij} + b_{ij} \quad (kA)_{ij} = k a_{ij}$$

Matrix Multiplication

For A of size $m \times p$ and B of size $p \times n$, the product AB is $m \times n$:

$$(AB)_{ij} = \sum_{k=1}^p a_{ik} b_{kj}$$

Row i of A dotted with column j of B .

Example: Compute AB where $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$.

$$AB = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

AQA Exam Tip: Matrix multiplication is NOT commutative: $AB \neq BA$ in general. Always check the order. AQA may test this explicitly.

Properties of Matrix Multiplication

- Associativity: $(AB)C = A(BC)$
- Distributivity: $A(B + C) = AB + AC$
- $(AB)^T = B^T A^T$
- $AI = IA = A$

3. Determinants

2x2 Determinant

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

3×3 Determinant (Cofactor Expansion)

Expanding along the first row:

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Example: Find $\det \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 1 & 2 & -2 \end{pmatrix}$.

$$\begin{aligned} &= 2[(-1)(-2) - (4)(2)] - 1[(0)(-2) - (4)(1)] + 3[(0)(2) - (-1)(1)] \\ &= 2[2 - 8] - 1[0 - 4] + 3[0 + 1] \\ &= 2(-6) - 1(-4) + 3(1) = -12 + 4 + 3 = -5 \end{aligned}$$

Properties of Determinants

- $\det(AB) = \det(A) \det(B)$
- $\det(A^T) = \det(A)$
- $\det(kA) = k^n \det(A)$ for an $n \times n$ matrix
- Swapping two rows changes the sign of the determinant
- If two rows are identical, $\det = 0$

4. Inverse Matrices

2×2 Inverse

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad \text{where } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \det A \neq 0$$

Example: Find the inverse of $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$.

$$\det A = 6 - 5 = 1$$

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$$

3×3 Inverse via Cofactors

The inverse of a 3×3 matrix A with $\det A \neq 0$ is:

$$A^{-1} = \frac{1}{\det A} (\text{adj } A) \quad \text{where } \text{adj } A = (C_{ij})^T$$

Here $C_{ij} = (-1)^{i+j} M_{ij}$ is the cofactor (signed minor) of element a_{ij} .

Example: Find A^{-1} for $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 0 \end{pmatrix}$.

$$\det A = 1(0 - 3) - 2(0 - 3) + 1(0 - 1) = -3 + 6 - 1 = 2$$

$$\begin{aligned} \text{Cofactors: } C_{11} &= +(0 - 3) = -3, C_{12} = -(0 - 3) = 3, C_{13} = +(0 - 1) = -1, \\ C_{21} &= -(0 - 1) = 1, C_{22} = +(0 - 1) = -1, C_{23} = -(1 - 2) = 1, \\ C_{31} &= +(6 - 1) = 5, C_{32} = -(3 - 0) = -3, C_{33} = +(1 - 0) = 1 \end{aligned}$$

$$\text{adj } A = \begin{pmatrix} -3 & 1 & 5 \\ 3 & -1 & -3 \\ -1 & 1 & 1 \end{pmatrix}, \text{ so } A^{-1} = \frac{1}{2} \begin{pmatrix} -3 & 1 & 5 \\ 3 & -1 & -3 \\ -1 & 1 & 1 \end{pmatrix}$$

AQA Exam Tip: The sign pattern for cofactors is $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$. Always verify your inverse by checking $AA^{-1} = I$ if time permits.

5. Solving Simultaneous Equations with Matrices

The system $A\mathbf{x} = \mathbf{b}$ has solution $\mathbf{x} = A^{-1}\mathbf{b}$ when $\det A \neq 0$.

Example: Solve $\begin{cases} 2x + y = 5 \\ x - y = 1 \end{cases}$.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \det A = -2 - 1 = -3$$

$$A^{-1} = -\frac{1}{3} \begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$$

$$\mathbf{x} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 6 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

So $x = 2, y = 1$.

Singular Matrices and System Analysis

When $\det A = 0$, the matrix is **singular** and has no inverse. The system $A\mathbf{x} = \mathbf{b}$ is either:

- **Inconsistent** — no solutions (parallel planes that don't all meet)
- **Consistent with infinitely many solutions** — the equations are dependent

AQA Exam Tip: For a 3×3 system, if $\det A = 0$, you must determine whether the system is inconsistent or has infinitely many solutions. Substitute a general value (e.g. let $z = t$) and check consistency of the remaining equations.

6. Linear Transformations in 2D

A 2×2 matrix A represents a linear transformation of the plane. The columns of A are the images of the standard basis vectors $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Standard 2D Transformations

Transformation	Matrix	Notes
Rotation by θ (anticlockwise about origin)	$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$	$\det = 1$
Reflection in x -axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$\det = -1$
Reflection in y -axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	$\det = -1$
Reflection in $y = x$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\det = -1$
Reflection in $y = -x$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	$\det = -1$
Enlargement, scale factor k	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$	$\det = k^2$
Stretch, factor k in x -direction	$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$	$\det = k$
Shear, factor k parallel to x -axis	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$	$\det = 1$

Area scale factor: Under the transformation represented by matrix A , areas scale by $|\det A|$. Orientation is preserved if $\det A > 0$ and reversed if $\det A < 0$.

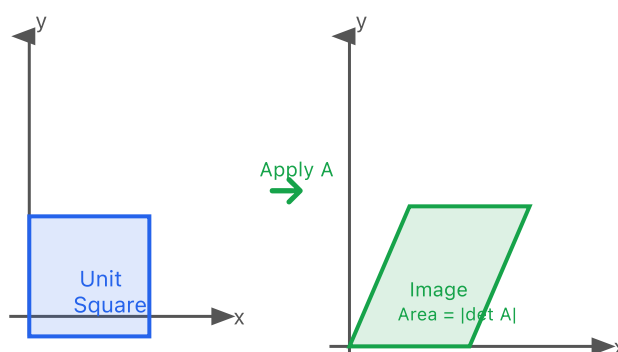
Composition of Transformations

Applying transformation B first, then transformation A : the combined matrix is AB (NOT BA). Read from right to left.

Example: The unit square has vertices at $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$. Under the transformation $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, find the image vertices and the area of the image.

Images: $\begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 3 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$\det A = 6 - 0 = 6$. Area of image = $|\det A| \times 1 = 6$.



7. Linear Transformations in 3D

A 3×3 matrix represents a linear transformation in 3D space. Key examples:

Transformation	Matrix
Rotation by θ about z -axis	$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Reflection in xy -plane ($z \rightarrow -z$)	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
Enlargement, scale factor k	$k I_3$

For 3D transformations, volumes scale by $|\det A|$.

8. Eigenvalues and Eigenvectors

Eigenvalue and eigenvector: A non-zero vector $\mathbf{v}$ is an *eigenvector* of matrix A with corresponding *eigenvalue* λ if:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Geometrically: the transformation A maps $\mathbf{v}$ to a scalar multiple of itself (same or opposite direction).

Finding Eigenvalues: Characteristic Equation

Rearranging: $(A - \lambda I)\mathbf{v} = \mathbf{0}$. For non-zero $\mathbf{v}$, the matrix $(A - \lambda I)$ must be singular:

$$\det(A - \lambda I) = 0 \quad (\text{characteristic equation})$$

Example: Find the eigenvalues of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

$$\det(A - \lambda I) = \det \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix} = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 2)(\lambda - 5) = 0 \Rightarrow \lambda = 2 \text{ or } \lambda = 5.$$

Finding Eigenvectors

For each eigenvalue λ , solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$.

Example (continued): Find eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

$$\text{For } \lambda = 2: (A - 2I)\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{0}$$

$$2x + y = 0 \Rightarrow y = -2x. \text{ Eigenvector: } \begin{pmatrix} 1 \\ -2 \end{pmatrix} \text{ (or any scalar multiple).}$$

For $\lambda = 5$: $(A - 5I)\mathbf{v} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{0}$
 $-x + y = 0 \Rightarrow y = x$. Eigenvector: $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

AQA Exam Tip: Eigenvectors are not unique — any non-zero scalar multiple is equally valid. AQA mark schemes accept any correct scalar multiple. State clearly which eigenvalue corresponds to which eigenvector.

Eigenvalues and Diagonalisation (Further Context)

If A has linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots$ with eigenvalues $\lambda_1, \lambda_2, \dots$, then forming $P = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots]$ gives $P^{-1}AP = D$ (diagonal), where $D = \text{diag}(\lambda_1, \lambda_2, \dots)$. This is used in solving certain differential equations and computing matrix powers.

9. Cayley-Hamilton Theorem

Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation. That is, if $\det(\lambda I - A) = \lambda^2 + p\lambda + q$ (for a 2×2 matrix), then $A^2 + pA + qI = O$.

Example: For $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$, the characteristic equation is $\lambda^2 - 7\lambda + 10 = 0$.

By Cayley-Hamilton: $A^2 - 7A + 10I = O$.

Verify: $A^2 = \begin{pmatrix} 18 & 7 \\ 14 & 11 \end{pmatrix}$, $7A = \begin{pmatrix} 28 & 7 \\ 14 & 21 \end{pmatrix}$
 $A^2 - 7A + 10I = \begin{pmatrix} 18 - 28 + 10 & 7 - 7 + 0 \\ 14 - 14 + 0 & 11 - 21 + 10 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \checkmark$

The Cayley-Hamilton theorem can be used to find A^{-1} or to express higher powers of A in terms of lower powers.

Using Cayley-Hamilton to find A^{-1} : From $A^2 - 7A + 10I = O$, rearranging:

$$10I = 7A - A^2 = A(7I - A)$$

$$A^{-1} = \frac{1}{10}(7I - A) = \frac{1}{10} \begin{pmatrix} 3 & -1 \\ -2 & 4 \end{pmatrix}$$

10. Summary of Key Results

Result	Formula / Statement
2x2 determinant	$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$
2x2 inverse	$A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$
$\det(AB)$	$\det(A) \det(B)$
Area scale factor	$ \det A $
Characteristic equation	$\det(A - \lambda I) = 0$
Eigenvector equation	$A\mathbf{v} = \lambda\mathbf{v}$
Cayley-Hamilton	Matrix satisfies its own characteristic equation
Rotation matrix	$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

11. Practice Checklist

- Can you multiply two 3×3 matrices without errors?
- Can you compute the determinant of a 3×3 matrix by cofactor expansion?
- Can you find the inverse of a 2×2 and 3×3 matrix?
- Can you use matrices to solve 3×3 simultaneous equations?
- Can you identify and write down the standard 2D transformation matrices?

- Can you find eigenvalues and eigenvectors of a 2×2 or 3×3 matrix?
- Can you apply the Cayley-Hamilton theorem to find A^{-1} or simplify matrix expressions?

Eigenvalues and Eigenvectors — Full 3×3 Worked Example

For a 3×3 matrix, the process is the same as 2×2 : find $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for each eigenvalue.

Example: Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

Step 1 — Characteristic equation:

$$\det(A - \lambda I) = \det \begin{pmatrix} 2 - \lambda & 1 & 0 \\ 1 & 2 - \lambda & 0 \\ 0 & 0 & 3 - \lambda \end{pmatrix}$$

$$\begin{aligned} \text{Expanding along the third column: } (3 - \lambda) \det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} &= (3 - \lambda)[(2 - \lambda)^2 - 1] \\ &= (3 - \lambda)(\lambda^2 - 4\lambda + 3) = (3 - \lambda)(\lambda - 1)(\lambda - 3) = -((\lambda - 3)^2(\lambda - 1)) \end{aligned}$$

Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 3$ (repeated).

Step 2 — Eigenvectors:

$$\lambda = 1: A - I = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}. \text{ Row reduce: gives } x + y = 0, z = 0. \text{ Eigenvector: } \mathbf{v}_1 =$$

$$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}. \lambda = 3: A - 3I = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \text{ Row reduce: gives } x = y, z \text{ free. Two linearly}$$

$$\text{independent eigenvectors: } \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Cayley-Hamilton Theorem

Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation. If $p(\lambda) = \det(\lambda I - A)$, then $p(A) = 0$ (the zero matrix).

Proof sketch (2×2): For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the characteristic polynomial is $p(\lambda) = \lambda^2 - (a + d)\lambda + (ad - bc)$. Compute $A^2 - (a + d)A + (ad - bc)I$ directly to verify it equals $\mathbf{0}$. [Full proof uses the adjugate matrix.]

Applications:

- Finding A^{-1} : if $A^2 - \text{tr}(A) \cdot A + \det(A) \cdot I = 0$, then $A^{-1} = \frac{1}{\det A}(\text{tr}(A) \cdot I - A)$.
- Reducing powers: express A^n as a linear combination of I and A (for 2×2) using the characteristic equation repeatedly.

Example: $A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$. Char poly: $\lambda^2 - 5\lambda + 4 = 0$. By Cayley-Hamilton: $A^2 - 5A + 4I = 0$, so $A^2 = 5A - 4I$. Then $A^3 = A \cdot A^2 = 5A^2 - 4A = 5(5A - 4I) - 4A = 21A - 20I$, etc.

Diagonalisation

If an $n \times n$ matrix A has n linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$, then A is **diagonalisable**:

$$P^{-1}AP = D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

where $P = [\mathbf{v}_1 | \mathbf{v}_2 | \dots | \mathbf{v}_n]$ (columns are eigenvectors).

Computing A^n : From $A = PDP^{-1}$ it follows that $A^n = PD^nP^{-1}$, and $D^n = \text{diag}(\lambda_1^n, \dots, \lambda_n^n)$ is trivial to compute.

Example: $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$. Eigenvalues $\lambda = 1, 3$; eigenvectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$A^n = P \begin{pmatrix} 1^n & 0 \\ 0 & 3^n \end{pmatrix} P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 3^n \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3^n - 1 \\ 0 & 3^n \end{pmatrix}$$

Geometric Interpretation of the Determinant

For a 2×2 matrix A :

- The **signed area** of the image of the unit square under A equals $\det(A)$.
- If $\det(A) > 0$, orientation is preserved; if $\det(A) < 0$, orientation is reversed (reflected).
- If $\det(A) = 0$, the transformation collapses the plane to a line or point ($\text{rank} \leq 1$).

For a 3×3 matrix, $|\det(A)|$ gives the factor by which volumes are scaled. The scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \det[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$ gives the signed volume of the parallelepiped.

AQA Exam Technique — Matrices:

- **Eigenvalues:** Always fully factorise the characteristic polynomial. For 3×3 , try $\lambda = \text{tr}(A)/3$ as a first guess, or look for a row that gives a factor.
- **Eigenvectors:** After finding λ , row-reduce $A - \lambda I$ carefully — set one variable as the parameter.
- **Cayley-Hamilton:** Show explicitly that $p(A) = \mathbf{0}$ by computing A^2 , then combining.
- **Diagonalisation:** State P , D , and confirm the order matches (column k of P corresponds to diagonal entry k of D).
- **Transformation composition:** ST means apply T first, then S . Read right to left.
- **Mark scheme:** Correct $A^{-1} [M1]$, correct solution $[A1]$; wrong method but correct determinant $[M1]$ if using Cramer's rule.

Standard Transformation Matrices (2D)

Transformation	Matrix	Notes
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Rotation by θ anticlockwise	$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$	$\det = 1$, preserves area
Reflection in x -axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$\det = -1$
Reflection in y -axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	$\det = -1$
Reflection in $y = x$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\det = -1$
Reflection in $y = -x$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	$\det = -1$
Reflection in $y = x \tan \alpha$	$\begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}$	$\det = -1$
Stretch by a in x , b in y	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$\det = ab$
Enlargement scale factor k	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$	$\det = k^2$
Shear: x -direction, factor k	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$	$\det = 1$

Solving 3×3 Systems — Row Reduction

Given $A\mathbf{x} = \mathbf{b}$, form the augmented matrix $[A|\mathbf{b}]$ and row reduce to echelon form.

Example: Solve $x + 2y + z = 4$, $2x + y - z = 1$, $3x - y + 2z = 5$.

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 1 & -1 & 1 \\ 3 & -1 & 2 & 5 \end{array} \right)$$

$R_2 \rightarrow R_2 - 2R_1: (0, -3, -3 | -7)$. $R_3 \rightarrow R_3 - 3R_1: (0, -7, -1 | -7)$. $R_3 \rightarrow 3R_3 - 7R_2:$

$(0, 0, 18|28)$. So $z = 14/9$. From R_2 : $-3y - 3(14/9) = -7 \Rightarrow y = 7/9$. From R_1 : $x = 4 - 2(7/9) - 14/9 = 4 - 28/9 = 8/9$. Solution: $x = 8/9, y = 7/9, z = 14/9$.

The Three Inconsistency Cases for 3×3 Systems

After row reduction, the augmented matrix reveals the nature of the solution:

Reduced form	Interpretation
Full rank 3: $\left(\begin{array}{ccc c} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{array}\right)$	Unique solution
Rank 2, consistent: bottom row $(0, 0, 0 0)$	Infinite solutions — line in $\mathbb{R}^3$
Rank 2, inconsistent: bottom row $(0, 0, 0 k)$ with $k \neq 0$	No solution (triangular prism)
Two rows proportional (rank 1 only): but inconsistent	No solution (parallel planes)

Properties of Eigenvalues

- **Trace:** $\text{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$ (sum of eigenvalues = sum of diagonal entries)
 $\det(A) = \lambda_1 \lambda_2 \dots \lambda_n$ (product of eigenvalues)
- **Powers:** If $A\mathbf{v} = \lambda\mathbf{v}$, then $A^k\mathbf{v} = \lambda^k\mathbf{v}$
- **Inverse:** If A is invertible with eigenvalue λ , then A^{-1} has eigenvalue $1/\lambda$
- **Transpose:** A and A^T have the same eigenvalues (but generally different eigenvectors)
- **Symmetric matrices:** Always have real eigenvalues and orthogonal eigenvectors

Using trace and determinant to find eigenvalues of a 2×2 : For $A = \begin{pmatrix} 5 & 2 \\ 1 & 4 \end{pmatrix}$: $\text{tr}(A) = 9$, $\det(A) = 20 - 2 = 18$. Characteristic equation: $\lambda^2 - 9\lambda + 18 = 0 \Rightarrow (\lambda - 3)(\lambda - 6) = 0$. So $\lambda_1 = 3, \lambda_2 = 6$. Check: $3 + 6 = 9 = \text{tr}(A) \checkmark, 3 \times 6 = 18 = \det(A) \checkmark$.

Worked Cayley-Hamilton Application

Find A^{-1} using Cayley-Hamilton for $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$. Characteristic polynomial: $\lambda^2 - 7\lambda + 10 = 0$. By Cayley-Hamilton: $A^2 - 7A + 10I = 0$. Rearranging: $10I = 7A - A^2 \Rightarrow I = \frac{1}{10}(7A - A^2) = A \cdot \frac{1}{10}(7I - A)$. Therefore: $A^{-1} = \frac{1}{10}(7I - A) = \frac{1}{10} \begin{pmatrix} 4 & -1 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 2/5 & -1/10 \\ -1/5 & 3/10 \end{pmatrix}$. Verify: $\det(A) = 12 - 2 = 10$, so $A^{-1} = \frac{1}{10} \begin{pmatrix} 4 & -1 \\ -2 & 3 \end{pmatrix} \checkmark$.

Matrix Multiplication — Properties and Examples

Key properties:

- Matrix multiplication is **not** commutative in general: $AB \neq BA$.
- Associative: $(AB)C = A(BC)$.
- Distributive: $A(B + C) = AB + AC$.
- $(AB)^T = B^T A^T$, $(AB)^{-1} = B^{-1} A^{-1}$.
- For square A : $\det(AB) = \det(A) \det(B)$.

Example: Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$. Find AB and BA to show they differ.

$$AB = \begin{pmatrix} 4 & 7 \\ 8 & 15 \end{pmatrix}, \quad BA = \begin{pmatrix} 3 & 4 \\ 11 & 16 \end{pmatrix}. \text{ Indeed } AB \neq BA.$$

Matrix Powers Using Diagonalisation

Example: $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$. Find A^n .

Eigenvalues: $\lambda = 3, \lambda = 2$.

$$\text{For } \lambda = 3: \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix} \mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

For $\lambda = 2$: $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, D = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}, P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

$$A^n = PD^nP^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3^n & 0 \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3^n & 3^n - 2^n \\ 0 & 2^n \end{pmatrix}.$$

Inverse of a 3×3 Matrix

For a 3×3 matrix A : $A^{-1} = \frac{1}{\det A} \text{adj}(A)$ where $\text{adj}(A)$ is the transpose of the matrix of cofactors.

Cofactor $C_{ij} = (-1)^{i+j} M_{ij}$ where M_{ij} is the minor (determinant of the 2×2 matrix with row i and column j removed).

Example: $A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}$.

$$\det A = 1(1 \cdot 1 - 3 \cdot 0) - 2((-1) \cdot 1 - 3 \cdot 2) + 0 = 1 + 14 = 15.$$

$$\text{Cofactors } C_{11} = 1, C_{12} = 7, C_{13} = -2, C_{21} = -2, C_{22} = 1, C_{23} = 4, C_{31} = 6, C_{32} = -3, C_{33} = 3.$$

$$\text{adj}(A) = \begin{pmatrix} 1 & -2 & 6 \\ 7 & 1 & -3 \\ -2 & 4 & 3 \end{pmatrix}, A^{-1} = \frac{1}{15} \begin{pmatrix} 1 & -2 & 6 \\ 7 & 1 & -3 \\ -2 & 4 & 3 \end{pmatrix}.$$

Linear Transformations — Composition

Applying transformation T_1 (matrix A) then T_2 (matrix B) is represented by BA (note the reversed order).

The image of point $\mathbf{x}$ under transformation T is $A\mathbf{x}$.

Example: Reflect in the x -axis ($A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$) then rotate 90° anticlockwise ($B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$).

Combined transformation:

$$BA = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \text{reflection in } y = x.$$

Example — Stretch and rotate: Scale by 2 then rotate 45° :

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}.$$

$$BA = \begin{pmatrix} \sqrt{2} & -\sqrt{2} \\ \sqrt{2} & \sqrt{2} \end{pmatrix}. \text{ This represents an enlargement by 2 combined with a rotation.}$$

Matrices — AQA Quick Reference

Transformation	Matrix
Rotation by θ (anticlockwise)	$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$
Reflection in $y = x \tan \alpha$	$\begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}$
Enlargement scale factor k	$k \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
Stretch in x by a , y by b	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$
Shear (parallel to x -axis)	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$

Characteristic Polynomial and Minimal Polynomial

The **characteristic polynomial** of A is $\chi_A(\lambda) = \det(\lambda I - A)$.

The **minimal polynomial** $m_A(\lambda)$ is the monic polynomial of lowest degree such that $m_A(A) = O$.

By Cayley-Hamilton: $\chi_A(A) = O$. The minimal polynomial divides the characteristic polynomial.

Example: For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Characteristic polynomial: $(\lambda - 1)^2$. Both eigenvalues are 1.

Is $m_A = (\lambda - 1)$? Test: $A - I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq O$. So minimal polynomial is $(\lambda - 1)^2$.

Example 2: For $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I$.

Characteristic polynomial: $(\lambda - 2)^2$. But $A - 2I = O$, so minimal polynomial is $(\lambda - 2)$.

AQA Matrices Exam Technique:

- For eigenvalue/eigenvector questions: always verify by computing $A\mathbf{v} = \lambda\mathbf{v}$ for your final answer.
- For Cayley-Hamilton: substitute A for λ in the characteristic equation — be careful with $\lambda^0 = I$ (not 1).
- To find A^{-1} via Cayley-Hamilton: rearrange so that $I = f(A)$ and read off A^{-1} .
- Diagonalisation: $A^n = PD^nP^{-1}$ only works when A has n linearly independent eigenvectors (i.e. is diagonalisable).
- For a 2×2 matrix: if $\text{tr}(A) = \lambda_1 + \lambda_2$ and $\det(A) = \lambda_1\lambda_2$, these give a quick check on your eigenvalues.
- Eigenvectors are not unique — any non-zero scalar multiple is also an eigenvector. In AQA, state the direction clearly or write the general form $k\mathbf{v}$.
- Row operations: when solving $(A - \lambda I)\mathbf{v} = \mathbf{0}$, use row reduction. The rows should be linearly dependent (since $\det = 0$), so one equation is redundant — use the remaining equation to express components in terms of a free parameter.

Determinant Properties — Reminder

- $\det(AB) = \det(A) \det(B)$
- $\det(A^{-1}) = 1/\det(A)$
- $\det(A^T) = \det(A)$
- $\det(kA) = k^n \det(A)$ for an $n \times n$ matrix
- Swapping two rows (or columns) multiplies the determinant by -1 .
- Adding a multiple of one row to another does not change the determinant.